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- **Black brackets []** indicate fields the editor fills in.
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 - If a section is noted as **optional** and is not needed, delete the section title and black text.

NENA [NG911 Management Considerations for EIDO Interoperability]

ABSTRACT: [one or two sentences describing the content of the document]

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The CRM team will provide the remainder of the information on this page.

NENA-INF-NNN.N.Na-YYYY

NENA Board of Directors Approval:
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Month DD, YYYY
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Prepared by:

National Emergency Number Association (NENA) [insert the applicable Committee, Subcommittee, and/or WG names]

NOTE to document editor: The Table of Contents defaults to showing three heading levels. If the Working Group determines that it is important to show a fourth heading level, customize the Table of Contents using the following instructions:

1. Click within the Table of Contents, and select the **References** tab.
2. Select **Custom Table of Contents** from the **Table of Contents** drop-down menu.
3. Set **Show Levels** to 4 under **General**.
4. Select **OK**.

Remove the **List of Tables** and **List of Figures** if they are not useful for the document.

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93 EXECUTIVE OVERVIEW

94 **NOTE to document editor:** Every document includes an Executive Overview. The Overview
95 should be unique and not just be cut-and-pasted from another section. In most cases, this
96 section should be no more than one (1) page and have no subsections. This section does not
97 contain normative language. It should provide a brief summary of the entire document which
98 can be developed by answering the following questions:

- 99 • What is the goal of the document/what problem or issue is it trying to solve?
- 100 • What is the background of the issue?
- 101 • Who is affected by the issue and the solution?
- 102 • Are there any other important things that readers of this document should know or be
103 aware of?

104 If additional information is required, the Working Group may choose to create a Background or
105 Introduction section at the beginning of the main body of the document. This can also be
106 included as a Purpose and Scope section or a Benefits section that may have existed in earlier
107 versions of the document's Executive Overview if the Working Group determines it is not
108 appropriate to continue to maintain these subsections here.

109 This document describes the management of the information ~~to governing the movement of~~
110 ~~Emergency Incident Data Objects (EIDOs)~~ incident data from one part of the Public Safety
111 Answering Point (PSAP) / Emergency Communications Center (ECC) to other parts of the
112 PSAP/ECC or to other PSAP/ECCs or ECCs using an Emergency Incident Data Object (EIDO).
113 The core scope of this document covers:

- 114 1. What is an EIDO and when is one sent;
- 115 2. What data can be put into an EIDO;
- 116 3. Who is authorized to send and receive an EIDO - criteria used to determine;
- 117 4. What best practices should be applied to EIDO management; and
- 118 5. Are interagency agreements necessary to exchange EIDOs outside of the NG911 system.

119 Data that is available today in evolving smart technologies is not always receivable by today's
120 PSAP/ECCs, and today's PSAP/ECC capabilities commonly are very limited in interoperable data
121 sharing except through manual or customized automated processes that are unique to each
122 PSAP/ECC. NG911 technology addresses these limitations through standardized functional
123 elements and interfaces which are policy driven application processes. Each agency's NG911
124 system will function based on the agency's data management policies. Currently NENA has no
125 standard or best practice for PSAP/ECC Managers and the local Authority Having Jurisdiction
126 (AHJ) in the development of their data management policies.

127 It is imperative that NG9-1-1 Authorities and PSAP/ECC Managers understand that data
128 management, through the policies they write, will drive their system's interoperability. They
129 must also understand the corresponding responsibilities, including risks and risk mitigation
130 through interagency agreements defined in policies, as well as security policies and processes.

131 Our approach in this document is to provide PSAP/ECC Managers and AHJ with an
132 understanding of NG911 incident data, the functional role of the Emergency Incident Data
133 Object referred to as EIDO, and the NG911 environment; along with recommendations for best
134 practices in policy development considerations as they impact the critical tasks of dispatch

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136 operations. Careful policies will be key to successful implementation of NG911 services and
137 their impact on the PSAP/ECC, interoperability with other PSAP/ECCs, while also addressing
138 critical data protection needs.

139 Through good data management a PSAP/ECC can redefine operational tasks which impact the
140 workload yet expand the level of emergency service they provide. Local responsibility must
141 address both data protection and improving service.

142 The scope ~~limits~~ of this document shall serve as a basis for PSAP/ECC management NG911
143 implementation and the sharing of data as it directly applies to managing the emergency
144 incident response. It is beyond the scope of this document to provide a standard for 9-1-1 data
145 sharing during large scale disaster recovery, among various Government services agencies,
146 with the media, or any non-public safety entities that aren't ancillary to the emergency incident.
147 While these services provide societal necessities, data sharing interoperability between the
148 PSAP/ECC and these services has been identified for future work. Many other NG911
149 considerations outside of the scope of this document can be found in related NENA, APCO, or
150 other Standards Developing Organizations (SDO) published documents.

151 **EXAMPLE:**

152 This document outlines the crucial aspects of managing tasks within the Public Safety
153 Answering Point (PSAP) environment, particularly as the industry transitions to Next Generation
154 9-1-1 (NG9-1-1). The document emphasizes managing critical tasks, addressing performance
155 issues, and preventing task overload. This is particularly important as new systems and
156 technologies are introduced with NG9-1-1.

157 PSAP managers are expected to handle staffing, retention, training, and quality assurance,
158 often with limited resources. These challenges are projected to intensify with the ongoing
159 migration to NG9-1-1. Stakeholders in 9-1-1 systems must understand the issues related to
160 task overload and how it affects service delivery. This understanding is crucial for effective
161 9-1-1 operations.

162 The document's recommendations are grounded in industry best practices, research, and
163 experience. They aim to improve the quality and delivery of 9-1-1 services.

164

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167 training, education, outreach, and advocacy. Our vision is a public made safer and more secure
168 through universally-available state-of-the-art 9-1-1 systems and better-trained 9-1-1
169 professionals. Learn more at <https://www.nena.org>.

170 This Information Document (INF) is published by the National Emergency Number Association
171 (NENA) as an information source for those that support emergency calls such as 9-1-1 System
172 Service Providers (9-1-1 SSPs), network interface vendors, system vendors,
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174 complete design or operation specifications or parameters or to assure the quality of
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177 limited to:

- 178
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 - Reflecting changes in the design of equipment, network interfaces, or services described
- 179
180
181 herein.

182 This document is an information source for the voluntary use of public safety communication
183 centers, such as 9-1-1 Public Safety Answering Points (PSAPs). It is not intended to be a
184 complete operational directive.

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188 information used. NENA recommends that readers contact their 9-1-1 Authority and 9-1-1 SSP
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190 compliance with current regulations.

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205 NENA information documents do not introduce new normative language as defined in RFC 2119
206 [2] and later updated by 8174 [3].

207 UNIT OF MEASUREMENT

208 **NOTE to document editor:** The NENA Development Group Operational Procedures, ADM-001,
209 Section 6.1 Appendix A—NENA Metric Policy states that Units of the International System (SI)
210 and/or United States Customary System (USCS or USC) may be included as appropriate. The
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212 identify the unit of measurement: US, Metric, both US and Metric, and Not applicable. If no
213 measurement is included in the requirement, the appropriate unit of measurement is "Not
214 applicable." The editor should select one of the following statements to include in the
215 requirement.

216 Where a measurement is found in the document, the unit of measurement is [US, Metric, or
217 both US and Metric]. When a measurement is used in the document, the unit of measurement
218 is identified.

219 No measurement is used within this requirement; therefore, the unit of measurement is "Not
220 applicable".

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222 **NOTE to document editor:** If the main body of your document includes Artifacts, include this
223 "Artifacts" section and the text immediately below; otherwise, delete this section and, if not
224 referenced elsewhere in the document, remove reference [4], [5], and [6] in Section 6,
225 References.

226 There are Artifacts [1] in this document, as defined in *NENA Rules for Use of Version Control*
227 *Repositories*, NENA-ADM-012 [4]. Artifacts defined in NENA documents, such as
228 Representational State Transfer (REST) interfaces and JavaScript Object Notation (JSON)
229 objects, usually use OpenAPI [5] as their formal description. In some cases, XML is used [6].

230 In the description of the interface or object in NENA documents, a table may appear that
231 describes the parameters to the interface and the data objects. These tables are informative;
232 the Artifact file is the normative documentation of the interface and data objects unless
233 otherwise specified in the document. If there is a discrepancy between the table and the
234 Artifact file, the Artifact file controls unless otherwise specified in the document.

235 In the tables, there is a column named "Required." The following three (3) values may occur in
236 this column:

- 237 • A "Yes" means the parameter or member MUST be present. It MUST appear in the
238 "required" section of the Artifact file.
- 239 • A "No" means that the parameter or member MAY be present. It MUST NOT appear in
240 the "required" section of the Artifact file.
- 241 • A "Conditional" means that the presence of the parameter or member is conditional. This
242 value alerts the reader that a business rule is specified, which controls the presence of
243 the parameter or member. It MUST NOT appear in the "required" section of the Artifact

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247 file. This is because notation formats like YAML do not provide a way to support a
248 conditional business rule.
249 Business rules are contained in this document for associated Artifacts. These business rules
250 MUST be followed in implementing the standard.

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269 **NOTE to document editor:** For all revisions to the initial document, the WG/editor will need to
270 provide a synopsis of the Reason for Issue/Reissue. While it should be as brief as possible, it
271 should permit the reader to have a good understanding of what has changed in each revision. If
272 a major rewrite was performed, a statement to that effect is appropriate instead of trying to list
273 all the changes. The CRM team will insert the official document number and Approval Date.

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276 version tags.

277 NENA reserves the right to revise this document. Upon revision, the reason(s) will be provided
278 in the table below.

Document Number	Approval Date	Reason For Issue/Reissue
NENA-STA-[NNN.1-YYYY]	[Month DD, YYYY]	Initial Document
NENA-STA-[NNN.2-YYYY]	[Month DD, YYYY]	[Example: Major rewrite; updated URLs]
NENA-STA-[NNN.2.1-YYYY]	[Month DD, YYYY]	[Example: Minor update; Added widget escape procedure]
NENA-STA-[NNN.2.1a-YYYY]	[Month DD, YYYY]	[Example: Fixed unintentional inconsistency in widget processing]

Document Number	Approval Date	Reason For Issue/Reissue
NENA-STA-[NNN.3-YYYY]	[Month DD, YYYY]	[Example: Modifications to section(s) (list the sections)]
NENA-STA-[NNN.4.1-YYYY]	[Month DD, YYYY]	[Example: Log event added to Logging Service in Section X.Y; authoritative version at https://xyz]
NENA-STA-[NNN.4.1a-YYYY]	[Month DD, YYYY]	[Example: corrections to schema in Appendix ABC; authoritative version at https://xyz]

1 EMERGENCY INCIDENT DATA OBJECT (EIDO) DEFINITION AND INTRODUCTION TO THE MANAGEMENT OF INTER-OPERABLE NG9-1-1 SYSTEMS

NOTE to document editor: Replace "Heading Title" here and in each section with an appropriate title that describes the subject matter of that section.

1.1 An Introduction to the EIDO

NOTE to document editor: Document topic-specific section/subsection headings and descriptive text are to be included in this section. The following instructions provide greater detail. However, if the editor has questions regarding how to use this template, they should contact the WG Chair(s), the Committee Chair(s), or the Committee Resource Management Team for assistance.

The Heading Title (that the authors of this document will provide above) serves as the beginning of the specific detailed portion of the document. The authors of this document would add any necessary headings within this section to complete this document. These sections can be as long as needed to support the document topic. Editors are advised to use as few heading levels as possible for readability, preferably not to exceed Heading Level 4 and to include no more than Heading Level 6. This section can also be broken up into multiple Heading 1 levels if the content organization is better served than having everything in one section that uses several nested heading levels.

In addition, the document editor should

- be consistent with the use of bullets and item numbering,
- identify heading levels for all sections and subsections by using the appropriate WORD Styles,
- use WORD's built-in "cross-reference" function to indicate where the reader can find associated "References" that are listed below in Section [6.6 References](#), or for other document navigation purposes, and
- upload the source files for all diagrams, images, and illustrations (e.g., Visio, Illustrator, PowerPoint) into the NENA Workspace for future editing.

Additional information on these and more editing pro-tips can be found in the [General Document Editing Help](#) section at the end of this Template.

In a NG9-1-1 environment Emergency Incident information is transferred between parties seamlessly through a standard interoperable method. The mechanism that enables this is the EIDO, which is a JSON-based (JavaScript Object Notation) data structure used to share emergency incident information between and among authorized entities and systems. Think of an EIDO as a current snapshot of the information known of an incident at the moment in time when it is sent to another entity based on your agency's data sharing policies. NENA has adopted the JSON-based EIDO for sharing incident information among authorized Next Generation 9-1-1 (NG9-1-1) entities and systems.

Throughout this document emergency incident information will be referenced as Incident Data. An incident is a real-world occurrence, or event, for which one or more calls may be received. It will be the responsibility of the Agency to determine Incident Data sharing policies related to which agencies it will share Incident Data with. Once data sharing policies are established, the agencies functional elements will be configured to reflect them and automatically share Incident Data.

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Commented [MH3]: May need to consider introducing the concept of subscribers in the introduction.

Commented [AH3R2]: The subscriber concept is discussed in the consideration section. Verify.

325 ~~A functional element is a set of interfaces and operations on those interfaces to accomplish a~~
 326 ~~task. Within i3 there are three primary functional elements: Call Handling, Incident Handling~~
 327 ~~and Dispatch. A Functional Element can subscribe to other Functional Elements, both internally~~
 328 ~~within an agency and externally to Functional Elements belonging to outside agencies. A~~
 329 ~~subscriber may subscribe to either all Incident EIDs or a subset thereof, a single incident's~~
 330 ~~EIDs or to all EIDs related to a specific resource. What information a subscriber can see is~~
 331 ~~dependent on the hosting agency's Data Right's Management (DRM) policies.~~
 332 ~~You may want to reference the Glossary definition of FE (Functional Element) for more detail.~~
 333 ~~[https://kb.nena.org/wiki/FE_\(Functional_Element\)](https://kb.nena.org/wiki/FE_(Functional_Element))~~
 334

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Commented [AH4]: Has this been covered in functional elements below. Consider deleting if redundant or moving if not. Also 297 and 298

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Commented [MH5R2]: Consider referencing the normative standard NG PSAP?

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335 1.2 History

336 In E9-1-1 systems, caller information is retrieved at a PSAP from an ALI (Automatic Location
 337 Identification) database. The call handling controller then delivers this information to CAD
 338 (Computer Aided Dispatch) systems over a serial interface described in NENA-STA-027.3-2018.
 339 The information in the ALI response and in the data transferred to the CAD system included
 340 caller information such as location, caller name, and responsible police, fire, and EMS services
 341 for that location.

342 NENA intends the use of EIDO to relay emergency incident data between functional elements
 343 within a PSAP, between PSAPs, between PSAPs and other external entities such as law
 344 enforcement agencies, fire departments and emergency medical services (EMS). Unlike legacy
 345 E9-1-1 systems, which require a telecommunicator to verbally notify agencies of changes in
 346 incident status, the standard EIDO format allows NG9-1-1 systems to notify agencies of
 347 incident status changes using machine driven automatic methods.

349 1.3 Applicable Functional Elements

350 In legacy PSAPs, we have two large systems we generically call "CPE" and "CAD". CPE handles
 351 the 9-1-1 call, and CAD handles incidents and the response to those incidents.

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352 Within a PSAP, all NG9-1-1 compliant systems that deal with incident data use EIDs as the
 353 standard way to express what one system or agency knows about an incident. By using EIDO
 354 as the common way to exchange incident data, we get a level of interoperability between
 355 disparate systems. The EIDO is the data format of the information the Call Handling FE provides
 356 the Incident Handling FE about a 9-1-1 call and the data format of the information the Incident
 357 Handling FE tells the Dispatch FE the data that was collected and entered by the call taker and
 358 received with the call.

359 To standardize the exchange of information in the computer based NG9-1-1 environment,
 360 Functional Elements have been defined. There are four Primary Functional Elements currently
 361 defined: Call Handling, Incident Handling, Dispatch and IDX (Incident Data eXchange).
 362 Multiple Functional Elements may be combined in a single vendor product or service or may be
 363 provided by different vendors. All Functional Elements should interoperate across various
 364 vendor products using EIDs.

365 IDX is a Functional Element that facilitates the exchange of EIDs among other Functional
 366 Elements both within and external to an agency. Since these exchanges may be both internal
 367 and external, agencies must be aware of how this functionality will impact data flow when

forming their policies on EIDO transactions. The IDX will implement the EIDO management policies, for external entities so the policy will need to consider the destination in any data management considerations, which is the primary focus of this information document.

1.4 What Data Can Be Put into an EIDO

The components of EIDOs is are populated using the JSON Schema as defined in NENA-STA-021. The data components of an EIDO Emergency Incident Data Object Data Components include, but are not limited to:

- Agency Data
- Split/Merge Data
- Link Data
- Incident Data
- Call Data
- Call Back Data
- Updated Call Back Number
- Dispatch Data
- Disposition Data
- Notes Data
- Person Data
- Vehicle Data
- Location Data
- Additional Data
- Emergency Resource Data
- Alarm and Sensor Data

For more details regarding the data components of EIDOs please refer to NENA-STA-021.

Historical data is stored in the logging service. All details, including event history, regarding the incident that were embedded in previous EIDOs will be maintained by the logging service. EIDOs represent the current state of an incident. Detailing the logging service is outside the scope of this document. Refer to NENA-STA-10.3b-20214 for further information on the logging service.

INSERT DIAGRAM IN SECTION 2 OF DEPENDENCIES ?

Note: IDX as well as the process for Non-Public Safety Entities has not yet been defined.

1.5 Exchanging Information in the NG9-1-1 Environment

2 (DEPENDENCIES)

2.1 Who is Authorized to Send and Receive EIDOs?

Agents within a PSAP as well as external agencies electronically subscribe to the Incident to maintain situational awareness of incident status. The PSAP should create a Data Rights Management (DRM) policy that defines what information will be provided to the various entities that subscribe to the incident, both internally and externally.

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Commented [DH8]: John Milburn: Move lines 378 thru 394 to a different section and condense if needed.

Commented [DH8R2]: John will come back to group with suggestion on different section.

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Commented [AH9]: Agents within a PSAP, as well as external agencies electronically subscribe to the incident to maintain situational awareness of incident status to the current state. The PSAP must create a Data Rights Management (DRM) policy that defines what information will be provided to the various entities that subscribe to the incident both internally and externally.

Deleted: must

410 As mentioned in previous sections, NG9-1-1 functional elements send EIDOs based on agency
411 policies.

412 Recommendation on the primary EIDO will guide the PSAP to reflect on their relationships, legal
413 agreements or set of policies determining the information carried by EIDOs exchanged between
414 agencies.

415 EIDO transmissions follow a subscribe/notify model, where a notifier provides information about
416 an incident to a subscriber.

417 A notifier executes policies that control what information is distributed to a subscriber. These
418 data sharing policies ("data policies") are data documents that are based on PSAP operational
419 processes and the requirements of each organization and any applicable jurisdictional laws.

420 ~~The subscriber of information shall establish internal policies on how to subscribe and the types~~
421 ~~of conditions or status updates being requested from the various agencies or entities based on~~
422 ~~filters.~~

423 The PSAP Credentialing Agency (PCA) issues certificates to public safety agencies that have
424 been vetted by the appropriate Registration Authority (RA). This certificate allows the NG9-1-1
425 system to analyze the trustworthiness of agencies that send subscription requests to other
426 agency incidents. For example, a mutual aid partner that has been issued a certificate by the
427 PCA may be provided more information than a certified agency that doesn't have a mutual aid
428 agreement. In a situation where the IDX receives a subscription request from a non-certified
429 agency then it will send the agency a subscription rejection if this is the response defined in the
430 DRM policy. The process for certifying Non-Public Safety entities has been identified for future
431 work.

432

433 **2.2 The Role of Data Rights Management Policy in the sending and receiving of**
434 **incident data in an EIDO.**

435 In Legacy 911, telecommunicators relate incident information to agencies and individuals based
436 on their roles, established information sharing policies, and applicable laws. While it's still the
437 responsibility of the 911 Authority to establish data sharing policies in NG9-1-1, the IDX relates
438 incident information automatically to subscribers based on how the 911 Authority has
439 configured data rights in its vendor's system(s). The EIDO data sharing policies are the rules
440 the software uses to make decisions regarding which subscribers are provided what data.

441 Such rules should be dependent on the level of trust the 911 Authority has in the subscriber as
442 well as disclosure restrictions and the subscriber's role(s). Further considerations will be
443 detailed in the following section. See 'Data Rights Management Diagram' for an overview of
444 how the level of trust a 911 Authority has in a subscriber should impact the amount of
445 information they provide.

446

447 EIDOs communicate changes in the state of an emergency incident between emergency
448 systems functional elements (FEs), public safety agencies and other entities. Agents and
449 telecommunicators will never work directly with EIDOs. They will take actions in the user
450 interface that trigger EIDOs to be transmitted. The information that is shared with a recipient
451 will vary based on the configuration of the data rights management policy and what kind of
452 request a recipient has made. A typical PSAP would also include subscriptions for key
453 stakeholders such as law, fire and EMS.

454

Commented [DH10]: Need a better understanding from i3 group and NG PSAP on programmable policies vs agency policies (Mike V raised for discussion)

Commented [DH10R2]:

Commented [DH11]: Confirm with Dan Mongrain or tech folks on next call

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Commented [DH12]: 09232024 - We cover basic credentialing in lines 349-356 of section 2.3.1. Do we want to move these lines and the level of trust diagram to this section? I'm fine either way but it's something for the group to consider. Reference section 4.6 security section

Commented [MH12R2]: 11/4/24 this is ok where it is, just add reference to section 4.6 for more details

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The 911 Authority will be required to configure data rights in vendor systems both for internal use within the PSAP and for external public safety stakeholders. While these may include neighboring PSAPs the authority has a mutual aid agreement with, these can also include PSAPs with which there are no formal agreements, including those outside the authority's region, state or province.

When configuring data rights in Agency applications, per NENA-STA-024.1.1-2025 "The data rights management policy MUST conform to local, state/provincial, and federal laws and regulations." Care should be taken to conform to laws related to criminal justice information and patient confidentiality like CJIS, CPIC and HIPAA such that a subscriber's discipline and role within an organization dictate what information is available; for example, only providing personal health related information to EMS. The 911 Authority will never modify the DRM policies associated with an EIDO it receives from another agency, nor will agencies receiving an EIDO sent by the 911 Authority be able to modify the DRM policy associated with it. Their software should automatically be configured to respect the other agencies' DRM policies by their vendor. The following chart is an example of a matrix to assist in conceptualizing and managing data sharing rules.

• **Matrix for Managing Data Sharing Rules**

Agency Type	PSAP (in area) Cooperating with Mutual Aid Agreements	PSAP (out of area) No Mutual Aid Agreement- Ex. Nearby – Same State/Province/ County	PSAP (out of area) No Mutual Aid Agreement Ex. – Out of State/Province/ County	Gov't Services	Large Scale Disaster Recovery	Non-Public Safety	Media
PSAP (in area)	Automated Policy (incident info yes/no) ☐ Unrestricted ☐ Confidential? ☐ Fire? ☐ Police? ☐ EMS?	Automated Policy (incident info yes/no) ☐ Unrestricted ☐ Confidential? ☐ Fire? ☐ Police? ☐ EMS?	Automated Policy (incident info yes/no) ☐ Unrestricted ☐ Confidential? ☐ Fire? ☐ Police? ☐ EMS?	Future work	Future work	Future work	Future work
PSAP (out of area)	Automated Policy (incident info yes/no) ☐ Unrestricted ☐ Confidential? ☐ Fire?	Automated Policy (incident info yes/no) ☐ Unrestricted ☐ Confidential? ☐ Fire?	Automated Policy (incident info yes/no) ☐ Unrestricted ☐ Confidential? ☐ Fire?	Future work	Future work	Future work	Future work

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Commented [DH15]: Do we start with Full Access, Limited Access, No Access? As an Appendix, include the chart from Matthew to fully understand all the capabilities.

Pick out what is applicable for Call Transfers.

Review Dan's presentation from ESWG – 5 steps – put it in the contribution folder. Slide 12

Getting initial feedback from agencies: put something together for NENA annual. Offer a survey before hand and review at conference.

Commented [DH16]: For discussion on 8/12/24 - should we include this as a separate column or leave off entirely.

	☐ Police? ☐ EMS?	☐ Police? ☐ EMS?	☐ Police? ☐ EMS?				
Government Service	Future work	Future work	Future work	Future work	Future work	Future work	Future work
Large Scale Disaster Recovery	Future work	Future work	Future work	Future work	Future work	Future work	Future work
Non-Public Safety	Future work	Future work	Future work	Future work	Future work	Future work	Future work
Media	Future work	Future work	Future work	Future work	Future work	Future work	Future work

Deleted: Traditional agency roles may vary for a limited time in a response. For example, first on scene may be law enforcement and the situation requires the officer to begin initiating first aid. In such a situation the responding officer may have a need-to-know health related information specific to an individual. If the 911 system is preconfigured to only release this kind of health-related information to EMS agencies, then the telecommunicator may need to convey this information to the responding officer orally. Alternatively the PSAP may choose to establish policies to deal with situations where preconfigured Data Rights Management solutions are inadequate.

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Commented [MH20]: This bullet is covered in lines 619-627. Only caveat is the last lines about Requests via Conveyance. We may need to mention that somewhere. We have not discussed prior to this point in the document, but have a number of instances in subsequent sections.

Commented [DH20R2]: Re-confirm with Dan Mongrain about 709 - 711 is this request for awareness handled in Conveyance or is it still future work.

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~~Agents and telecommunicators will never work directly with EIDOs. They will take actions in the user interface that trigger EIDOs to be transmitted. The information that is shared with a recipient will vary based on the configuration of the data rights management policy and what kind of request a recipient has made.~~

Policy Decision Considerations:

— is this a request for awareness or BOLO memo scenario? For specific incident use request for service in conveyance. Maybe requests go through standard mechanisms and EIDO's follow per standard and policy store.

2.3 Security of Incident Data

The incident information shared via the IDX and other functional elements will adapt dynamically based on the subscribing agency's role and changes in incident status. Within the PSAP environment, Federal, State, Province, Territory and local government laws and regulatory requirements should govern sensitive data transmission across fire, EMS, and law enforcement. Public safety stakeholders will require access to less redacted data, while agencies outside of public safety might receive only targeted information relevant to their specific resource request(s).

Inter-PSAP communication for mutual aid shall prioritize essential data points such as incident location, nature, and required resources. ~~withholding additional incident data until the resource request is accepted by the receiving agency.~~ For agencies with frequent collaboration, real-time

updates on resource locations and availability will be crucial. Sharing of incident data shall seamlessly integrate with existing provisioning processes and communication workflows within each PSAP. Maintaining a common format for incident data streamlines dispatcher workload and facilitates a common operating picture between PSAPs and responders.

A typical PSAP should include subscriptions for key stakeholders: law enforcement, fire, EMS, towing services, utilities, and highway departments. While emergency service agencies will receive incident information based on the Data Rights Management policies established by their PSAP with variations based on their roles, support agencies like utilities and highway departments will primarily receive location and incident details essential to their response.

Subscriptions for emergency services shall employ an automated flow process driven by CAD recommendations. Dispatchers can readily add frequently used support agencies as resources within the CAD system.

Subscription requests should be streamlined through pre-configured workflows and initiated during mutual aid requests or incident transfers. Crucially, both requesting and accepting agencies must adhere to their respective policies and all applicable laws. The Law Enforcement Data Exchange, for example, must comply with Canada's regulatory framework related to CPIC and state/territory regulations, while medical information sharing in the United States will strictly adhere to HIPAA privacy and security rules.

To safeguard sensitive data, subscription requests shall follow a phased approach. Initial requests for mutual aid will provide only essential information (nature, location, resources). If the receiving PSAP cannot fulfill the request, sensitive data remains protected. Upon acceptance, incident data flows freely for the incident's duration, tailored to each agency's function.

↓

▲

3 (IMPACTS AND CONSIDERATIONS)

NOTE to document editor: This section is optional. If this section is not needed, delete the section title and the text.

Use this section to convey any additional information that the Working Group believes is important for the reader to understand. Examples of impacts and considerations include the following:

- operational
- technical
- security-related
- level of effort
- time required to implement
- cost or cost-recovery
- any other information the group believes the reader should understand when using this document

The organization of this section and any subsections is at the discretion of the editor and the Working Group.

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If your jurisdiction doesn't already have a DRM policy in place, then use section 4 of NENA-STA-040.2-2024 as a baseline for a minimum DRM security policy. If you've already established a DRM policy with mutual aid partners in accordance with the NG-SEC standard, you can use this as a starting point for a larger policy for both internal and external agencies and roles.

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Commented [DH27]: Need a better understanding from i3 group and NG PSAP on programmable policies vs agency policies (Mike V raised for discussion)

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3.1.1 Policy Considerations for EIDO Sharing

~~Local agencies SHALL establish clear policies governing access to incident information based on agency role and data types. For example, a policy might stipulate that all law enforcement agencies responding to an incident receive comprehensive incident data until released by the incident commander. However, to protect potential investigations, policies should restrict access for other disciplines. This should involve limiting data visibility within 911 systems and utilizing encrypted communication channels for law enforcement.~~

~~To place the [above matrix](#) in context, consider that all the data components of an EIDO listed in 1.4 MUST have DRM policies defined in your NG9-1-1 vendor system(s) down to the level of each individual element of each data component. The system will automatically send incident updates to subscribers based on these policies. As an example, consider the Notes Data Component which may contain notes taken by emergency service agents and responders on scene at an incident. EIDO's containing this data component could easily contain information which, by law, has associated disclosure limitations. How your PSAP defines DRM policies in your vendor system(s) defines who receives what information in an EIDO.~~

~~Some basic questions you might ask yourself when defining your DRM policies may include... Is the information sensitive? Do you have a mutual aid agreement with the subscribing agency? What's the subscriber's role (Law, Fire, EMS, PSAP)? How far away are they from your agency? How much do you trust the subscriber? The amount of information you're willing to provide a mutual aid partner may vary significantly from what you're willing to provide to an agency two hundred miles away but you provide the more distant subscriber with 'something' because they're trusted by the PCA. The matrix is a guideline for general considerations you should take into account when defining your DRM policies.~~

~~That said, traditional agency roles may vary for a limited time in an incident response. For example, first on-scene may be law enforcement, and the situation requires an officer to begin initiating first aid. In such a situation the responding officer may have a need-to-know health related information specific to an individual. If the 911 system is preconfigured to only release this kind of health-related information to EMS agencies, then the telecommunicator may need to convey this information to the responding officer orally. Alternatively, the PSAP may choose to establish policies to deal with situations where preconfigured Data Rights Management solutions are inadequate. Such consideration may be necessary because as previously mentioned, in the NG9-1-1 environment vendor systems SHALL be configured to automatically send incident data based on your agency's DRM policy. The system won't know when to make exceptions.~~

~~The Data Rights Management Diagram below helps to summarize the general data sharing levels a PSAP might have based on the role and level of trust the PSAP has in the subscriber.~~

~~NG9-1-1 systems necessitate granular policies that define data access for each subscriber role and the level of trust the policyholder has in the subscribing agency. These policies should categorize the data components outlined in NENA-STA-021 down to the element level data types (e.g., medical, warrant, scene safety) and incorporate all relevant legal and security regulations. By automating "if-then" decision-making processes, these policies alleviate the manual burden currently placed on dispatchers.~~

PSAPs already have established contingency policies for situations during system overflows and tied up resources. These plans provide a roadmap to defining what data is exchanged during

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emergencies. So, by using your existing relationships as a framework, develop your data sharing policies in a similar way being cognizant of data privacy and stewardship. Assuming you have a good relationship with at least one other PSAP leverage that asset so that you both understand how you will subscribe to and exchange data adhering to policies you have created.

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This shift towards automated incident data transmission policies necessitates a heightened focus on security. Robust s Security measures must be implemented to protect sensitive information transmitted and received via EIDO throughout the incident lifecycle. At the end of the day, you are as good as your Data Rights Management policy. NENA suggests working with existing support agencies in your jurisdiction such as your NG9-1-1 system vendor(s), your Information Technology Department (or equivalent), Human Resources and legal team to see if they've already established a data security policy for your jurisdiction. If existing policies generally conform to the NENA Security for Next Generation 9-1-1 Standard (NG-SEC) (NENA-STA-040.2-2024) then work within your existing framework to establish your EIDO DRM policies.

If your jurisdiction doesn't already have a DRM policy in place, then use section 4 of NENA-STA-040.2-2024 as a baseline for a minimum DRM security policy. If you've already established a DRM policy with mutual aid partners in accordance with the NG-SEC standard, you can use this as a starting point for a larger policy for both internal and external agencies and roles.

Commented [AH30]: From 4/10/27 suggestions. ACCEPTED 4/27/26

3.1.2 Transitioning from Human-Driven to Automated Policies

While many automated data sharing policies should mirror existing human-driven practices, those subject to frequent change (due to staffing fluctuations, social factors, etc.) may need to remain under human control. In some situations, vendors may need to create systems in such a way that 911 administration can effectively 'disable' the automatic

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transmission of incident data. For example, if a high-ranking government official is involved in a vehicle crash, a 911 admin may wish to highly restrict information related to that incident. However, by transitioning stable, rule-based policies into automated workflows, PSAPs can significantly enhance efficiency and reduce the cognitive load on dispatchers. There is also the issue of exposure to information and imagery that some find disturbing especially with the nature of this work. It is recommended that guidelines from the above standard are implemented as soon as practicable.

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Commented [AH31R2]: NENA STA-052.1-202x NENA Standard for Incident Related Imagery (IRI) This needs updated to reflect that.

Commented [AH32]: This will be replaced with a synopsis Call to Action to reinforce that there needs to be solidifying of policies. 4/27/26

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3.1.3 Transition to Security Policies

This shift towards automated data policies necessitates a heightened focus on security. Robust security measures must be implemented to protect sensitive information throughout the incident lifecycle. At the end of the day, you are as good as your Data Rights Management policy. We suggest reading NENA Security for Next Generation 9-1-1 Standard (NG-SEC) specifically section 4 which deals with policy management. This gives good guidance on implementing security policy from the management perspective. Below are some specific security considerations to address the aforementioned.

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Historically, decisions about who could see incident information lived outside the 9-1-1 system, i.e., in memoranda of understanding, mutual-aid agreements, and local standard operating procedures, with telecommunications and supervisors applying those rules manually, one incident and/or request at a time. In the NG9-1-1/i3 environment, that same intent is expressed digitally as Data Rights Management (DRM): element-level rules, defined per

683 subscriber role and level of trust, that the vendor system enforces automatically and that are
684 governed through the PCA subscription model. The substance of your existing agreements
685 doesn't disappear, it is translated from prose and handshake into machine-enforceable policy.

686 The DRM policy itself - translating your existing rules into XACML(eXtensible Access Control
687 Markup Language) a standard that provides a policy-based framework for defining and
688 enforcing authorization rules, determining who can access specific resources and under what
689 conditions.

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690 In the legacy environment, your data-sharing rules lived in people's heads, in MOUs, and in
691 SOPs, and a telecommunicator applied judgment at the moment of release. In i3, those rules
692 are expressed in XACML v2.0, the OASIS standard that NENA i3 (NENA-STA-010.3, §5.6)
693 adopts as the mechanism for DRM and that the NGCS and PSAP systems enforce automatically.
694 You will not normally hand-write XACML, your vendor's policy administration tooling generates
695 it and holds it in the Policy Store. However, you are responsible for getting the logic right and
696 stating it precisely enough that a machine can apply it with no person in the loop.

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697 The most important mental shift is to default-deny. A person could improvise and share
698 something reasonable in a situation the rules never anticipated; XACML does the opposite,
699 anything you do not explicitly permit is withheld. So the work is not to write down what you
700 want to protect, it is to enumerate exactly what you are willing to expose, to whom, and under
701 what circumstances. Everything else is denied by default.

Commented [DH34]: XACML v2.0 is an open OASIS standard defining an XML-based language and processing model for creating and evaluating fine-grained, attribute-based access control (ABAC) policies across enterprise systems. **XACML v2.0** is an open OASIS standard defining an XML-based language and processing model for creating and evaluating fine-grained, attribute-based access control (ABAC) policies across enterprise systems.

702 To get there, take each of your existing policies, your MOUs, mutual-aid understandings, and
703 local release practices, and decompose them into the four questions XACML asks of every
704 request:

<https://www.oasis-open.org/standard/xacmlv2-0/>

705 • Who is asking (Subject). The subscribing entity, identified by its PCA-traceable
706 credentials: agency role (Law, Fire, EMS, PSAP), the specific agency, mutual-aid status, and the
707 level of trust you assign it. This is where the role-and-trust groupings from the matrix in the
708 previous section become concrete, enforceable attributes.

709 • What data (Resource). The specific EIDO data components and, critically, the individual
710 elements within them, classified per NENA-STA-021, location, incident type, person/medical,
711 vehicle, warrant, scene-safety, and so on. "Share the incident" is not specific enough; the policy
712 must say which elements.

713

714 • What operation (Action). Typically, to receive or subscribe to that element as the
715 incident updates.

716 • Under what circumstances (Environment / Conditions). Context such as incident type,
717 whether the incident commander has released information, the subscriber's relationship or
718 distance, and whether the subscription is scoped to a single incident ID or is a standing
719 qualified-filter subscription.

720 Each rule then resolves to a Permit or Deny, optionally carrying Obligations, actions the system
721 must take when it enforces the decision, such as redacting a field, logging the disclosure, or
722 flagging that the information may need to be conveyed by other means. Obligations are how
723 you carry over the "yes, but..." clauses of your existing policy.

724 The practical method, and the deliverable your vendor and developers will need from you is to
725 write each rule in plain language first, validate it against your sharing matrix, and only then

have it encoded in the Policy Store. For example: "Permit EMS-role subscribers dispatched to this incident to receive medical elements; deny medical elements to all other roles." Working in plain English keeps the policy auditable by the people who actually own the data, and it surfaces conflicts before they are buried in XML.

Two issues deserve deliberate attention as the set grows. First, conflicts: when more than one rule could apply to the same request, XACML resolves them with a combining algorithm, and for life-safety data a deny-overrides posture, any applicable deny wins and is the conservative choice. Decide this on purpose rather than inheriting a vendor default. Second, exceptions: as noted earlier, the system cannot improvise — it releases exactly what the policy says and nothing more. Encode the foreseeable exceptions as environment conditions (for instance, an incident-commander release flag) and pair the automated policy with a documented manual path for the situations no rule anticipated.

Finally, treat the policy set as living. Validate it against representative scenarios before go-live to confirm it exposes only what you intend, review your role-based access at least annually as NG-SEC requires, and revise the rules as roles, partners, and statutes change.

The security of the EIDO as it moves

Confidentiality, integrity, and authenticated transport, addressed by the Conveyance standard (NENA-STA-024) and i3 transport protections such as TLS using PCA-traceable credentials.

Your organization's overarching security governance

Acceptable use, access control, data classification, incident response, and the rest of the documented program a PSAP maintains. This is the domain of the NENA Security for Next Generation 9-1-1 Standard (NG-SEC, NENA-STA-040.2-2024). NG-SEC describes how to stand up and run that program from a management perspective; it is the enterprise context your DRM policies live inside, not a substitute for them.

3.2 Security of EIDO Data

EIDO data encompasses a wide range of information, from statuses of resources, Personally Identifiable Information (PII), HIPAA protected information, to sensitive information respective to an open investigation. With advances in technology all data is susceptible to the growing capabilities of eavesdroppers and attackers. The Next Generation (NG9-1-1) systems, applications, and flow process security must develop best practices for the safety of both the data and ultimately the responders and greater community. NENA-STA-024.1.1-2025, *Conveyance of Emergency Incident Data Objects (EIDOs) between Next Generation (NG9-1-1) Systems and Applications* addresses the security for the EIDO during its life within technology components. Remaining security risks that must be addressed lie within the human factor. And whereas, in previous 9-1-1 operations, the risk was always present, even exacerbated by growing staffing turnover rates. In NG9-1-1 those risks can be mitigated to a large degree through the subscription policies addressing Data Rights Management, developed by the PCA. Today agencies are cautious of data risks to the point of hindering interoperability. As subscriber policies are established for the protection of data, interoperability and regional resource allocation could become a commonality.

Commented [DH35]: James Kinney's submission 6/8/26

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Deleted: <#>InterPSAP Communications—A detailed review of a given PSAP's data-protection policy is advised. Also, and understanding of data retention and destruction policies so that you can record and account for EIDO's that were shared. (see 4.1.4.3 Data Owner in the ¶ NG-SEC doc)¶
A Data Custodian should be identified with PSAP's and agencies with which a given PSAP would be entrusting their EIDO's sensitive data with. (see 4.1.4.4 Data Owner in the ¶ NG-SEC doc)¶
Finally, as stated it is likely you have at least one relationship with a neighboring PSAP. Use the policies you have developed already with them as you guidance for digital data sharing. Reviewing each other's policy and reviewing. (see 4.2.2.5 Data Owner in the ¶ NG-SEC doc)¶

Commented [DH36]: Amy H content

Commented [HM36R2]: Amy will expand to cover curiosity requires and CJIS and Rick will look for Fire related security information.

Commented [HM36R3]: 15:51:55 From Pierre Foucault To Everyone:
- CANUTEC it is
15:55:15 From Rick Erickson To Everyone:
- Reacted to "CANUTEC it is" with 🙌

Commented [AH37]: Updated to new version. 6/20/25

Commented [AH38]: Do we want to look at the security information earlier in the document that mentions PCA moved down here to keep it all together. If we leave it in the beginning we may want to insert a reference to this section.

Commented [AH38R2]: The first mention is in line 384.

Commented [DH39]: John Milburn will suggest new edits.

Commented [MH39R2]: 11-8-24 Does this need to stay or should it be deleted.

Commented [DH39R3]: Groups states these statements should stay.

Commented [AH39R4]: RESOLVED 3/10/25

Commented [AH39R5]:

We suggest reading NENA Security for Next Generation 9-1-1 Standard (NG-SEC) specifically section 4 which deals with policy management. This gives good guidance on implementing security policy from the management perspective. Below are some specific security considerations to address the aforementioned.

Remaining security risks that must be addressed lie within the human factor. In Legacy 9-1-1 operations, the risk was always present and is exacerbated by growing staff turnover rates. In NG9-1-1 these risks can be mitigated to a large degree through the policies addressing Data Rights Management which are implemented by the agency. An agency can create groupings, such as those detailed in the matrix in the previous section, with predefined data rights based on role(s).

~~NENA STA-024.1a-2023 (2.10 Security) "All transports specified in this document are protected by TLS using credential (for those entities within an ESInet) traceable to the PSAP Credentialing Agency (PCA)". We are saving government services, non-public safety entities and media for future work so detailing security for these entities may be out of scope for this version of the document. For the security of EIDO creation reference the Conveyance and JSON Documents as appropriate.~~

For Best Practices for EIDO's we want to apply these criteria (and then see if Conveyance can verify?)

Agency Roles

Agent Roles

3.3 What best practices should be applied to EIDO Management

Recommendations based on anticipated issues and concerns:

Make sure you can transfer to agencies you know nothing about and receive from those you know nothing about as well.

You would want a policy for every legal agreement you have or one policy that covers them all. (agency policy vs software policy).

Subscriptions — software approved vs. human approved. When it's an agency you are not familiar with, you will want to know who is subscribed. Who wants to know and why?

Include definitions and what goes where....how data rights management fits into the policy store, policy routing.

Default — to reject a subscription because it's not a call they transferred to you.

Definition of legitimate will vary.

Querying loggers use case

1. — NG911 functional elements
2. — Public safety applications within the PSAP
3. — Originating PSAPs and assisting PSAPs
4. — Emergency responder applications

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Commented [MH40R2]: James is reviewing and will provide feedback on 6/8/26

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Commented [AH41]: Needs moved to first place we use PCA.

Commented [DH42]: Tackle on 10/21/2024 meeting to rephrase and state that there should be a back up plan. Required to have TLS or it won't work - here's why and make alternative arrangements.

Commented [MH42R2]: 11/4/2024 decision made to incorporate this into 4.6 as it is the only way to communicate at this time. No options are currently available.

Commented [DH42R3]: Group agrees to leave this here.

Commented [AH43]: Needs to be a citation.

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825 5. — Other law enforcement, fire, and EMS entities

826 Incorporate some of this language into the best practices section as appropriate:

827 Small PSAPs may not have much IT support. As moving to NG9-1-1, make sure you have IT

828 support to make the transition correctly. You must prepare or you'll have multiple kinds of new

829 software systems, you need an appropriate network, connection to neighboring agencies, etc.

830 Have conversations with them in the preparation phase. Don't be afraid to have a meeting with

831 all your vendors in one room — it helps to make sure everyone is on the same page, and to

832 prepare for the ultimate integrations.

833 Then look at your policies and procedures. You have procedures to do all of this manually. What

834 data is shared with other entities? Translate to data sharing policies. Write it all out. RACI =

835 Responsibility, Accountability, Consult, Inform Matrix. Make decision trees and flow charts to

836 make sure you and your partners and vendors understand what you're doing and how you plan

837 to get to your goal(s). Other methods include using sticky notes with lines or strings showing

838 the interdependencies. All of this sets the stage for positive change management, which will

839 help you all make a smooth transition.

840 Start slow and build on that when creating your management policies. Focus on what you're

841 doing today and figure out how to replicate and automate that.

842 Dan talks about rules for sharing data — you own your data. You can't share data you got from

843 another agency, but you can share their link to the data and a third party can request the data

844 via the link — the owner decides what to give the third party. The PCA certificates ensure that

845 each party knows who the other party is and the data owner can use its own policy rules to

846 decide what it will share with the requesting party.

847

848 [As EIDO is not currently widely available in production environments, we believe best practices](#)

849 [will continue to emerge with future deployments.](#)

850

851 [It's important to understand that with the adoption of EIDOs as the standard format for incident](#)

852 [data transmission and receipt that your data sharing relationship with other agencies is no](#)

853 [longer limited to those you have a mutual aid agreement with. Interagency agreements are](#)

854 [not necessary to exchange EIDOs between functional elements of different agencies. Each](#)

855 [agency must have rules in their Data Rights Management \(DRM\) policy to cover exchanges](#)

856 between agencies, including agencies unknown at policy writing time. Such agencies may be

857 outside your region, state/province or country.

858

859 [There is](#) no requirement to provide [an unfamiliar subscriber](#) with the same amount of incident

860 data you [would](#) provide a known public safety partner, but they [are](#) trusted by the PCA, so you

861 provide them with something regardless of whether you know them or not.

862

863 If you receive a subscription request from an entity that hasn't been issued credentials by the

864 PCA you have no obligation to provide them with anything. If your policy is to reject their

865 subscription request, then be sure to document this to your policy and verify your systems are

866 configured to reflect this.

867

868 3.4 Inter-Agency Agreements

869 Traditionally Interagency agreements, i.e. Mutual Aid Agreements, have been between entities

870 with an established relationship. The established relationship provided both entities were aware

871 of the available equipment and capabilities. With these arrangements details such as

872 reimbursement, insurance, method of contact and time required to respond were agreed upon

873 when agreements were signed. With the implementation of the EIDO the traditional

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Commented [DH44]: 6/15/26 Review with
Canadians to see how their experience has been.

Commented [DH44R2]: Approved by the group
6.15.26

interagency agreements may continue to be implemented between entities for established relationships; with additional agreements needing to be considered for the new scope of resources and capabilities entities may be aware of through EIDO implementation.

Agencies may need to refine their agreements to cover which EIDO data fields they will share with a particular agency. Agreements will continue, but will migrate from existing communication models to the EIDO models. Since NG911 will include non-traditional CAD to CAD partners, Agencies will need defined policies of which data to share those partners.

~~Interagency agreements are not necessary to exchange EIDOs between functional elements of different agencies. Each agency must have rules in the data sharing policy to cover exchanges between agencies, including agencies not known at policy writing time.~~

891

4 (RECOMMENDATIONS FOR ADDITIONAL DEVELOPMENT WORK)

NOTE to document editor: This section is optional. If this section is not needed, delete the section title and the text.

Recommendations for additional development work generally include information that should be considered in a future version of the document or is intended to be referred to another NDG Committee, Working Group, or Standards Development Organization. In general, additional development work can be considered as a list of important topics that the Working Group recommends for further development work.

The following list is intended to help the Working Group decide what to include as additional development work.

- Information that is valuable to the topic, although deemed out-of-scope for the current version's Charter. These items can be considered in a future version of the document, and many times will support the planning of the document's next version.
- Information or a topic that the Working Group recommends be considered by a different NDG Committee or Working Group, or Standards Development Organization (i.e., IETF, ATIS, APCO). If known, include the NDG Committee or Working Group, or the Standards Development Organization in the item description.
- Any information added to this section should be as complete as possible due to the future nature of the efforts. The current Working Group should not assume that it will be the Working Group assigned to support the future work.

Additional development work, whether as a revision to the document or a referral, is supported by the Issue Submission Form (ISF) process for consideration of the NDG membership and direction by the DSC. More information on the ISF process can be found by visiting https://www.nena.org/page/Submit_IssueNDG.

Upon publication, the Working Group Co-Chairs will review this section with their Committee Co-Chairs to determine the next step for each item listed in this section. This would include sending recommendations to external organizations or generating ISFs for NDG consideration.

This document references existing NENA [Standards (STA), Requirements (REQ), Information (INF), Reference (REF)]. Additional work may be required as noted below.

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Commented [DH45]: DO NOT DELETE!!! Moved to 3.3

Commented [AH45R2]: Unable to locate NENA-STA referenced.
John Milburn 4/10/26

Commented [JM45R3]: Moved to 3.3

Commented [DH45R4]: Group approves on 6.15.26

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921 **NOTE to document editor:** The section can be structured with numbered section headings, or
922 the table below can be used to document future work. Choose one format and remove the other
923 from this section.

924 **Suggested Heading format:**

925 4.1 [Future Work Item] Push

926 **NOTE to document editor:** Provide a distinct name for the future work item.

927 New Input: Details will be left to future versions of the document.

928 Version 1 will only have subscriptions. Areas identified for future work:

929 Are we entitled to recommend what should be the primary EIDO information for each type of
930 protocols?

931 (Push)

- 932 · Request for Service
- 933 · Request for Resource
- 934 · Request for Awareness

935 Pushes (EIDO are created when certain criteria is met, example triggers)

936 Conveyances (Request for Service, Request for Resources and Request for Awareness are all
937 forms of Conveyance which relay a function needed by an Agency)

938 Document what we do today – how do people know what they can/can't share over the phone

939 Define the "If this, then that" statements (IFTTT). Start with Mutual Aid Agreements. Ops
940 folks bring an example to next meeting.

941 Future work as related to EDXL-RM, in development in the conveyance group

942 4.2 [Description] What can be sent?

943 **NOTE to document editor:** Describe the future work item, its background, a justification for
944 why it should be done, and if applicable, the document section number.

- 945 • Requests for service
 - 946 ○ Call information on calls disconnected prior to transfer
 - 947 ○ Assistance for response from neighboring agency
 - 948 ○ Death notifications
 - 949 ○ Panic alarms
 - 950 ○ Aircraft, boat, or train incident
- 951 • Requests for resources
 - 952 ○ Bomb squad
 - 953 ○ K9
 - 954 ○ Ambulance

Commented [DH46]: When you get a request that includes an Incident ID, it should be handled instantly...based on policies. Wont' have time for a person to be consulted. Dependent on policy store.

Qual Filter – when a subscription is not for a particular incident. When based on a certain set of characteristics for an particular area. Might require manual intervention.

NENA Policy Routing Rules doc

Commented [DH47]: Action

Write plain English descriptions of what we would want developers to apply.

Commented [DH48]: Start here 5/17/21

Commented [AH48R2]:

- 955 ○ Law enforcement response
- 956 ○ Haz-Mat
- 957 ○ Power company/gas company
- 958 ○ Internal call outs
- 959 • Requests for awareness
 - 960 ○ BOLOs
 - 961 ○ ATLS
 - 962 ○ Emergency Alerts to hospitals
 - 963 ○ Hospital notifications of active diversion
 - 964 ○ Road closure information
 - 965 ○ Power outage and gas leak information
 - 966 ○ Extreme weather information, i.e. flooded areas
 - 967 ○ Power plant emergency alerts
 - 968 ○ FEMA/Emergency Management alerts
 - 969 ○ Video, photos, and other media
 - 970 ▪ Driver's license photos
 - 971 ▪ Social media photos
- 972 • When utilizing EIDO's you must understand the roles and Information relayed by not
 - 973 only the EIDO but also the conveyance protocol used. The three types of protocols are
 - 974 Request for Service, Request for Resource and Request for Awareness. The EIDO itself
 - 975 will provide call and/or incident information but the protocol by which the EIDO was
 - 976 delivered will provide incident information also. The reason you must consider the
 - 977 difference is that when you are deciding on what information will be in an EIDO you
 - 978 must understand what effect the conveyance of the EIDO will have. For example, if the
 - 979 EIDO uses the conveyance protocol of Request for Service, the location should be
 - 980 included in the EIDO body. If the conveyance protocol of Request for Resource is used,
 - 981 the EIDO should include the type of incident. An EIDO using a Request for Awareness
 - 982 may not require an ETA of responding units or apparatus.
- 983 • Another major difference between EIDO Data fields and Conveyance protocol is the
 - 984 acknowledge and response required by protocols. These responses convey more than
 - 985 just machine language of passing messages. By acknowledgement of message being
 - 986 received, the PSAP doesn't have to wonder or anticipate whether the information was
 - 987 relayed. If the EIDO was conveyed and acknowledged but a response not received then
 - 988 depending on the urgency of the Request, the PSAP may need to make another Request
 - 989 of another ECC. If an acknowledgement was not delivered back, then the ECC knows to
 - 990 try and make human contact or move on.

991

992 4.2.1 [Recommended Assignment]

993 **NOTE to document editor:** Include a recommended assignment such as a future version of
 994 this document, NDG Committee, Working Group, a new Working Group, or external Standards
 995 Development Organization (i.e., IETF, ATIS, APCO). If unknown, work with the Committee Co-
 996 Chairs to identify an appropriate assignment.

997 **Suggested Table format:**

998 **Table 4-1 Future Work for Consideration**

Recommended Assignment	Future Work
[Recommended Assignment]	[Future Work Item] [Description of future work item, with section number if applicable]
[Recommended Assignment]	[Future Work Item] [Description of future work item, with section number if applicable]

999 **5 ABBREVIATIONS, TERMS, AND DEFINITIONS**

Use this table to identify all terms that are unique to NENA or the 9-1-1 system or are important to this document. All abbreviations used in this document are to be included in this table. If only used once within the document, consider fully spelling out the term without its abbreviation so it does not need to be added to this table. Creating abbreviations that are not subsequently used in the document is not recommended.

Compare your definitions with the current [NENAbk Glossary](#) to avoid unnecessarily creating conflicting descriptions of the same term. **Review terms just before each approval ballot since the NENAbk Glossary is continually updated and definitions may have changed.**

Please add terms and abbreviations to the table using the following format:

- Place term or abbreviation alphabetically in the **Term or Abbreviation (Expansion)** column. For abbreviations, add the expansion (the spelled-out words) in parentheses, e.g., FCC (Federal Communications Commission).
- Provide a brief definition or description for the abbreviation or term.
- In the **Definition** column, you may also identify other Relevant NENA Documents or External References that provide greater detail about the term or abbreviation. URLs added to cited references will not be included in the NENAbk Glossary.

Instructions for Recommending Changes to the NENAbk (NENA Knowledge Base) Glossary

Use the third column to recommend changes to the NENAbk Glossary or confirm acceptance of the Glossary's definition in this document. The CRM team will remove the third column before the approved document is published. These options are available in the third column.

Select Recommendation drop-down box:

- **In Glossary**—The term and the definition in the document are the same as in the NENAbk Glossary. Please use the definition in the NENAbk Glossary.
- **Add**—The term does not exist in the NENAbk Glossary and meets the [criteria for inclusion](#). You may also identify other Relevant NENA Documents that provide greater detail about the term or abbreviation.
- **Delete**—The term exists in the NENAbk Glossary but is no longer needed or does not meet the [criteria for inclusion](#).
- **Don't Add**—The term is not in the NENAbk Glossary and does not meet the [criteria for inclusion](#) **OR** the WG needs a unique definition in this document other than what is in the NENAbk Glossary.
- **Modify**—Update the NENAbk Glossary with this term and definition. If the term originated in a different NENA document, consult with that document's Working Group (or its Committee if not currently active) to develop a single, agreed-upon definition.
- **Other**—Use this option to suggest other actions such as adding this term as a "Related Term" or "Also Known As" term to the definition of another term in the Glossary.
- **Insert Explanation**—If "Other" is selected **OR** a definition was modified in collaboration with another WG, **OR** any other explanation is needed, enter the details in the Insert Explanation text box (3rd column). Or else, delete the Insert Explanation text box.

1001 See the NENA Knowledge Base (NENAb) Glossary [1] for a list of terms and abbreviations used
1002 in NENA documents. Abbreviations and terms used in this document are listed below with their
1003 definitions as they apply to this version of the document. Please be aware that terms and
1004 definitions in the NENAb Glossary may be different than what is shown in this document.

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAb Glossary
		Select Recommendation Insert Explanation
LS (Location Server)	General term for the entity responsible for obtaining the location of the User Equipment (UE). See 3GPP TS 23.167	Add Add as a related term under IMS.
NGCS (Next Generation 9-1-1 Core Services)	The set of services needed to process a 9-1-1 call on an ESIInet. It includes, but is not limited to, the ESRP, ECRF, LVF, BCF, Bridge, Policy Store, Logging Services, and typical IP services such as DNS and DHCP. The term NG9-1-1 Core Services includes the services and not the network on which they operate. See Emergency Services IP Network.	Modify Definition modified in collaboration with the i3 Architecture WG
SHAKEN (Signature-based Handling of Asserted Information Using toKENS)	An industry framework for managing the deployment of Secure Telephone Identity (STI) technologies with the purpose of providing end-to-end cryptographic authentication and verification of the telephone identity and other information in an Internet Protocol (IP)-based service provider voice network. See ATIS-1000074-E	Add

1005 6 REFERENCES

1006 **NOTE to document editor:** Remove reference [4], [5], and [6] if the [Describing Artifacts](#)
1007 section was removed from the beginning of the document and it is not referenced elsewhere in
1008 the document.

- 1009 [1] National Emergency Number Association. "NENA Knowledge Base Glossary." Updated
1010 [insert the date indicated at "Approved updates to NENA Knowledge Base Glossary last
1011 made" as found at the cited link in the format Month DD, YYYY].
1012 <https://kb.nena.org/wiki/Category:Glossary>.
- 1013 [2] Internet Engineering Task Force. *Key words for use in RFCs to Indicate Requirement Levels*. S. Bradner. [RFC 2119](#), March 1997.
- 1014 [3] Internet Engineering Task Force. *Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words*. B. Leiba. [RFC 8174](#), May 2017.
- 1015 [4] National Emergency Number Association. *NENA Rules for Use of Version Control Repositories*. [NENA-ADM-012.1-2022](#). Alexandria, VA: NENA, approved November 8,
1016
1017 2022.
1018
1019

- 1020 [5] OpenAPI Initiative. *OpenAPI Specification 3.1.0 Released*.
1021 <https://www.openapis.org/blog/2021/02/18/openapi-specification-3-1-released>,
1022 February 18, 2021.
- 1023 [6] World Wide Web Consortium (W3C). *W3C XML Schema Definition Language (XSD) 1.1*
1024 *Part 1: Structures*. S. Gao, C. M. Sperberg-McQueen, and H. Thompson.
1025 <https://www.w3.org/TR/xmlschema11-1/>, April 5, 2012.
- 1026 **NOTE to document editor:** All sources used in the development of the document must be
1027 cited within the body text of the document. The reference citation is constructed in accordance
1028 with the Chicago Manual of Style (CMOS) (examples are provided below).
- 1029 When citing a NENA document, do not include an errata version letter in the document number
1030 and use the NENA Standards page URL (<https://www.nena.org/page/standards>) as the link to
1031 the document. Links to material outside of this document should be limited to the web page
1032 where the referenced document may be downloaded, as opposed to a link directly to the
1033 specific document. Questions regarding formatting references should be directed to the
1034 Committee Resource Management team at crm@nena.org.
- 1035 [7] [National Emergency Number Association. *NENA i3 Standard for Next Generation 9-1-1*.
1036 [NENA-STA-010.3-2021](#). Alexandria, VA: NENA, approved July 12, 2021.]
- 1037 [8] [National Emergency Number Association. *Emergency Incident Data Object (EIDO)*.
1038 [NENA-STA-021.1.1-202Y](#). Alexandria, VA: NENA (forthcoming).]
- 1039 [9] [Internet Engineering Task Force. *Session Initiation Protocol*. J. Rosenberg, H.
1040 Schulzrinne, G. Camarillo, A. Johnston, J. Peterson, R. Sparks, M. Handley, and E.
1041 Schooler. [RFC 3261](#), June 2002.]
- 1042 [10] [Association of Public-Safety Communications Officials International. *APCO International*
1043 *Core Competencies and Minimum Training Standards for Public Safety Communications*
1044 *Quality Assurance Evaluators (QAE)*. [APCO ANS 3.106.2-2017](#). Daytona Beach, FL:
1045 APCO, approved September 12, 2017.]
- 1046 [11] [International Telecommunications Union. *The international public telecommunication*
1047 *numbering plan*. [Recommendation E.164 \(11/10\)](#), November 18, 2010.]
- 1048 [12] [National Institute of Standards and Technology. "Cybersecurity Framework." Accessed
1049 December 6, 2024. <https://www.nist.gov/cyberframework>.]
- 1050 [13] [Trafficking Victims Protection Act of 2017, 34 U.S.C. § 10101 (2018),
1051 <https://www.congress.gov/115/plaws/publ393/PLAW-115publ393.pdf>.]
- 1052 [14] [U.S. Department of Justice, Federal Bureau of Investigation. *Criminal Justice*
1053 *Information Services (CJIS) Security Policy*. [CJISD-ITS-DOC-08140-5.8](#). Washington,
1054 DC: USDOJ, June 1, 2019.]
- 1055 [15] World Wide Web Consortium (W3C). *W3C XML Schema Definition Language (XSD) 1.1*
1056 *Part 1: Structures*. S. Gao, C. M. Sperberg-McQueen, and H. Thompson.
1057 <https://www.w3.org/TR/xmlschema11-1/>, April 5, 2012.
1058
- 1059 **NOTE to document editor:** Within the body text of the document, a numerical reference to
1060 the citation is provided at the end of the applicable text using the Word cross-reference
1061 function. The first occurrence of a cited work shall list the italicized name, followed by its
1062 document number (if provided) without a version number or errata letter. For example: *NENA*
1063 *i3 Standard for Next Generation 9-1-1*, NENA-STA-010 [5]. Subsequent citations of this
1064 document would be formatted using one of the following options:
- 1065 • *NENA i3 Standard for Next Generation 9-1-1* [5]
 - 1066 • NENA-STA-010 [5]

1067 For References cited within the body text of the document and where it is important that the
1068 reader only go to the "specific version" of the targeted document, the Reference should include
1069 the targeted document's version number.

1070 It is not advisable to include specific locations "within" a targeted document, such as a section
1071 number, paragraph number, or line number. Those specific locations are more likely to change
1072 when the targeted document is updated, even if the text referred to did not change.

1073

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1074 **7 APPENDICES**

1075 **7.1 Appendix X—[Appendix Title]**

1076 **NOTE to document editor:** The Appendices section is an optional section intended to include
1077 additional information referenced in your document. If this section is not needed, delete the
1078 section title and the text.

1079 Each Appendix should begin on a new page. Name each Appendix as Appendix A, Appendix B,
1080 etc.

1081

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1082 ACKNOWLEDGMENTS

1083 The National Emergency Number Association (NENA) [insert the applicable Committee,
1084 Subcommittee, and/or WG names] [developed, revised, reaffirmed] this document.

1085 NENA recognizes the following industry experts and their affiliated organization for their
1086 contributions to the [development, revision, reaffirmation] of this document.

1087 **NOTE to document editor:** The table below shall contain the names of the persons who have
1088 **actively contributed (verbally or in writing)** to the development of the document as
1089 determined by the Working Group Co-Chairs. The employer organization shall be the name of
1090 the person's affiliated organization, or "Individual" if not affiliated with an organization, at the
1091 time the document is approved by the NENA Board of Directors. Where appropriate, include a
1092 member's ENP or GISP status. Members should review their entry in this table before
1093 publication.

Members	Organization
Committee Co-Chair Name, XXX Committee Co-Chair	
Committee Co-Chair Name, XXX Committee Co-Chair	
WG Leader Name, Working Group Co-Chair	
WG Leader Name, Working Group Co-Chair	
WG Member Name, Technical Editor [if different than WG Leader]	
WG Member Name, Note Taker [if different than WG Leader]	
WG Member Name	
WG Member Name	
WG Member Name	
WG Member Name	
WG Member Name	

1094 SPECIAL ACKNOWLEDGMENTS

1095 **NOTE to document editor:** If a staff member is assigned to a specific working group for
1096 administrative support, the CRM team will adjust the text as needed. If there are questions
1097 about use of the *Additional names and description of reasons for acknowledgement* line, please
1098 contact the CRM team.

1099 NENA's Committee Resource Management Team has facilitated the production of this document
1100 through the prescribed approval process.

1101 [Additional names and description of the reasons for acknowledgment]

1102 The [Working Group Name] Working Group is part of the NENA Development Group that is led
1103 by:

- 1104 • Wendi Rooney, ENP, and Lisa Dodson, ENP, Development Steering Council Co-Chairs

- 1105
- 1106
- 1107
- Brandon Abley, ENP, Chief Technology Officer
 - April Heinze, ENP, VP, Chief of 9-1-1 Operations

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1108 GENERAL DOCUMENT EDITING HELP

1109 **NOTE to document editor:** NENA documents follow the Chicago Manual of Style (CMOS).
1110 Document editors are to use the following guidelines when editing NENA documents. If there
1111 are any questions regarding this section, contact the CRM team.

1112 Delete this entire section from the document before submitting the document for external
1113 review.

1114 Navigating the Styles Gallery and Styles Pane

1115 **NOTE:** Document editors MUST use the styles listed in the **Styles Gallery** in this template.
1116 These predefined Word styles are set to follow NENA's required font and formatting options.

- 1117 • The most frequently used styles are shown in the **Styles Gallery** on the **Home** tab.
- 1118 • Open the **Styles Panel** to see additional styles available in the document by clicking on
1119 the small arrow in the bottom right corner of the **Styles Gallery**.
- 1120 • To apply a style, select the text to be formatted, and then click on the desired style in
1121 the **Style Gallery** or **Styles Panel**.
- 1122 • Use the **Font** options on the **Home** tab to apply formatting (e.g., bold, italics).

1123 Watermark

- 1124 • All pages must include a **DRAFT** watermark.

1125 Font

- 1126 • **Verdana 10 pt**
- 1127 • When copying in text from a document, email, chat box, or other source, use the **Keep**
1128 **Text Only** paste option so text is added with the correct font and formatting.

1129 Paragraph Formatting

- 1130 • Use the Word Style **Body Text** for text in the document's main body. This adds visual
1131 space between paragraphs, eliminating the need for a blank line.

1132 Messages/Code in Text

- 1133 • **Font:** Lucida Console 10pt
- 1134 • **Left indent:** .5
- 1135 • **XML elements:** shown in brackets (<...>)
- 1136 • **JSON objects:** shown in double quotes with brackets ({ "... " })
- 1137 • **Values:** shown in double quotes ("...")
- 1138 • **Examples:**

```
1139 <note>
1140     <to>Dern Handy</to>
1141     <from>Brandon</from>
1142     <heading>Reminder</heading>
1143     <body>Don't forget to book!</body>
1144 </note>
1145
1146 {
1147     "iFoo":12,
1148     "sBar":"blurdybloop"
1149 }
1150
1151 "urn:service:sos"
```

1152 Footnotes

- 1153 • Cited documents¹, websites, and other references MUST be cited in the References
- 1154 section, not in footnotes. If something is important enough to include as a footnote, it is
- 1155 strongly recommended that the information be included in the document text.
- 1156 • **Numbers:** Superscript Verdana 7pt
- 1157 • **Text:** Verdana 8pt

1158 Punctuation

- 1159 • A comma is placed before a conjunction (e.g., and, or) in a series of three or more.
- 1160 • Periods are followed by a **single** space.
- 1161 • Periods and commas are placed **within** closing quotation marks.
- 1162 ○ **Exception:** If file names, words, or other strings to be typed are within quotation
- 1163 marks, any punctuation that belongs to the surrounding sentence should appear
- 1164 outside the quotation marks.
- 1165 • Colons, semicolons, question marks, and exclamation points are placed **outside** closing
- 1166 quotation marks (unless they belong within the quoted matter).
- 1167 • CMOS prefers square brackets [] within parentheses instead of parentheses within
- 1168 parentheses.
- 1169 • The abbreviations **i.e.** ("that is") and **e.g.** ("for example") are placed within parentheses,
- 1170 where they are followed by a comma.
- 1171 • CMOS prefers to limit the abbreviation **etc.** ("and other things") to parentheses. If used
- 1172 in a sentence, **etc.** is preceded by a comma. Do NOT use **etc.** at the end of a list that
- 1173 begins with *for example, such as, e.g.,* and the like.

1174 Non-breaking Hyphens

- 1175 • The CRM Team will insert non-breaking hyphens for 9-1-1 and all hyphenated words at
- 1176 time of publication. Editors should use regular hyphens during document development to
- 1177 avoid copy/paste of non-breaking hyphens being changed to spaces (no hyphen).
- 1178 • **alt+shift+hyphen** inserts a non-breaking hyphen which is treated as a letter, keeping
- 1179 the word on a single line.

1180 Vertical Lists—bulleted vs. numbered

- 1181 • Numbered lists are used to indicate the order in which tasks should be done, to suggest
- 1182 chronology or relative importance among the items, or to facilitate text references.
- 1183 • A vertical list is best introduced by a grammatically complete sentence, followed by a
- 1184 colon.
- 1185 • All items should be styled with the same syntax, either sentences or sentence
- 1186 fragments.
- 1187 • All items should use the same verb tense.
- 1188 • For bulleted lists, unless the items themselves form complete sentences, use lowercase
- 1189 for the first letter of each item (unless it is a proper noun) and omit periods.
- 1190 • For bulleted lists whose items require more prominence, capitalization may be preferred;
- 1191 choose one approach and follow it consistently.
- 1192 • Capitalize items in a numbered list even if the items do not consist of complete
- 1193 sentences. Closing punctuation is used only if items consist of complete sentences.
- 1194 • If the items in a vertical list complete a sentence that began in the introductory text,
- 1195 commas or semicolons may be used between the items (use semicolons if the items
- 1196 include internal punctuation), a period should follow the final item, and no colon is used
- 1197 in the introductory sentence. A conjunction (e.g., and, or) before the final item is
- 1198 optional.

¹ Example text to show footnotes with 7pt font for the footnote number and 8pt font for the footnote text.

- 1199 • Items in a vertical list should be short and to the point.

1200 Vertical List Symbology and Indentation

- 1201 • **Hanging Indents:** All use .25"
- 1202 • **Paragraph Indents and Spacing:**
 - 1203 ○ Don't add space between paragraphs of the same style (check box)
- 1204 • **Paragraph Line and Page Breaks:**
 - 1205 ○ Widow/Orphan Control (keep box checked) *Note: This only affects a bulleted paragraph consisting of 2-3 lines. Bullets with a single line are not affected.
 - 1206 ○ Keep with next (uncheck box)

1208 Bulleted and numbered lists can be formatted by using the **Format Painter** tool to apply the
1209 formatting below to the text in the core document. They can also be created by using the styles
1210 listed in the **Styles** gallery on the **Home** tab or by using the **Bullets** and **Numbering**
1211 formatting options within the **Paragraph** group on the **Home** tab.

1212 Bulleted List

- 1213 • Left Indent .25
- 1214 • Left Indent .25
 - 1215 ○ Left Indent .75
 - 1216 ○ Left Indent .75
 - 1217 ▪ Left Indent 1.25
 - 1218 ▪ Left Indent 1.25
 - 1219 • Left Indent 1.75
 - 1220 • Left Indent 1.75

1221 Numbered List

- 1222 1. Left Indent .25
- 1223 2. Left Indent .25
 - 1224 a. Left Indent .75
 - 1225 b. Left Indent .75
 - 1226 i. Left Indent 1.25
 - 1227 ii. Left Indent 1.25
 - 1228 1. Left Indent 1.75
 - 1229 2. Left Indent 1.75

1230 Alpha List

- 1231 A. Left Indent .25
- 1232 B. Left Indent .25 (AA-ZZ: Hanging indent .38")
 - 1233 a. Left Indent .75
 - 1234 b. Left Indent .75
 - 1235 i. Left Indent 1.25
 - 1236 ii. Left Indent 1.25
 - 1237 1. Left Indent 1.75
 - 1238 2. Left Indent 1.75

1239 Tables

1240 The example tables below MUST be copied and pasted into the document and adjusted as
1241 needed.

- 1242 • **Table Alignment:** Centered on page
- 1243 • **Table Width:** 6.75" (maximum)
- 1244 • **Table Caption:** Centered above table, Verdana 10pt Bold, Keep with Next pagination

- **Cell Margins:** R/L: 0.08"; Top: 0"; Bottom: 0.04"
- **Table Text Wrapping:** Around→Positioning→ Bottom 0.2"
- **Header Row:** Bold, repeats every page

To update the Table Number to reflect the correct section it was copied into, highlight the table number, right-click on the number, and select **Update Field**.

Table 7-1 Example Table with 3 Headings (3 columns)

Table 7-2 Example Table with 4 Headings (4 columns)

Figures

Figures are typically inserted between paragraphs or are placed at the end of a numbered section. Figure numbers should update automatically as they are inserted into the document, but if needed, they can be manually updated. To do so, highlight the figure number, right-click on the number, and select **Update Field**.

Insert figures on their own line using the following **Layout** settings:

- **Text Wrapping:** Top and bottom
- **Size:** Use **Absolute** settings with 6.75" maximum width
- **Position**
 - **Horizontal:** Alignment Centered relative to column
 - **Vertical:** Absolute position 0" below Line
 - **Options:** Check **Move object with text** and **Lock anchor**

Add captions once the figures are properly placed and sized. If figure size or position is adjusted later, captions may need to be redone for proper spacing between the caption and figure. Use the following **Caption** settings:

- **Label:** Figure
- **Position:** Below selected item
- **Style:** Caption (Verdana 10pt, bold)

Alt Text: Add a brief sentence to describe the diagram for someone who is blind or low vision. Example Alt text has been added to the figures below.

Copyrighted figure from an external source: Permission must be granted by the owner and may require a "used with permission" statement.

Store source files on NENA Workspace: Upload source files for all diagrams, images, and illustrations (e.g., Visio, Illustrator, PowerPoint) into NENA Workspace for future editing.

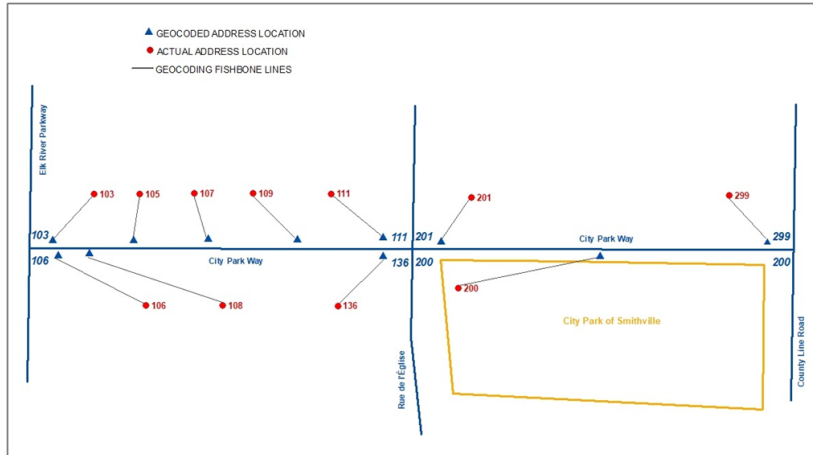


Figure 7-1 Effect of Geocoding Addresses against Actual Address Ranges

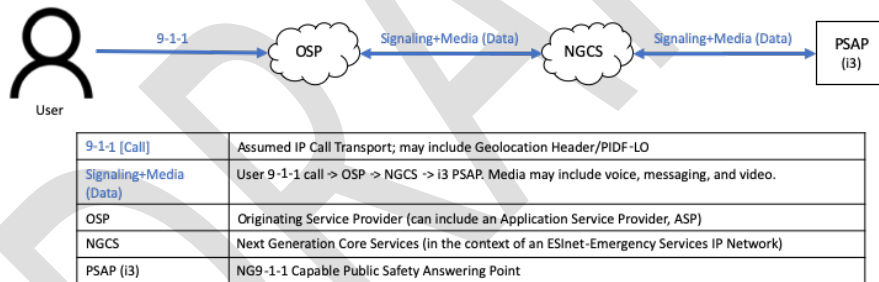


Figure 7-2 NENA i3 NG9-1-1 Reference Architecture

Cross-References—How to Jump Between the Citations

- Use <Ctrl+click> on a cross-reference to jump to the cited reference in the References section or to the targeted section, table, or figure **Cross-References – How to Jump Between** in the document.
- To return to the clicked cross-reference, use <Alt+Left Arrow key>.

Cross-References—How to Add

To add a Cross-reference **to a cited document, web page, or other cited item listed in the References section**, use these steps to insert the Reference number:

- 1286 • Insert the cursor after the cited reference in the text and on the **References** tab, select
- 1287 **Cross-reference**.
- 1288 • To add the cited reference number, in the **Cross-reference** pop-up box, set **Reference**
- 1289 **type:** to **Numbered item**, and set **Insert reference to:** to **Paragraph number**.
- 1290 • Scroll through the **For which numbered item:** list and select the appropriate Reference
- 1291 that is being cited.
- 1292 • Select the **Insert** button.
- 1293 • Change the font color of the inserted reference, including the brackets, to **Blue**.

1294 To add a Cross-reference **to a cited section** in the document, use these steps to insert the

1295 number and title of the section:

- 1296 • Insert the cursor where you want to cite the section in the text, and on the **References**
- 1297 tab, select **Cross-reference**.
- 1298 • In the **Cross-reference** pop-up box, set **Reference type:** to **Heading**, and set **Insert**
- 1299 **reference to:** to **Heading Number**.
- 1300 • Scroll through the **For which heading:** list and select the appropriate section heading
- 1301 that is being cited.
- 1302 • Select the **Insert** button.
- 1303 • Add a space to separate the section heading number from the section heading title.
- 1304 • In the **Cross-reference** pop-up box, set **Insert reference to:** to **Heading text**.
- 1305 • In the **For which heading:** list, select the appropriate section heading that is being
- 1306 cited.
- 1307 • Select the **Insert** button.
- 1308 • Change the font color of the inserted number and title of the section to **Blue**.

1309 To add a Cross-reference **to a cited table** in the document, use these steps to insert the

1310 number and title of the table:

- 1311 • Insert the cursor where you want to cite the table in the text, and on the **References**
- 1312 tab, select **Cross-reference**.
- 1313 • In the **Cross-reference** pop-up box, set **Reference type:** to **Table**, and set **Insert**
- 1314 **reference to:** to **Entire caption**.
- 1315 • Scroll through the **For which caption:** list and select the appropriate table that is being
- 1316 cited.
- 1317 • Select the **Insert** button.
- 1318 • Change the font color of the inserted table caption to **Blue**.

1319 To add a Cross-reference **to a cited figure** in the document, use these steps to insert the

1320 number and title of the figure:

- 1321 • Insert the cursor where you want to cite the figure in the text, and on the **References**
- 1322 tab, select **Cross-reference**.
- 1323 • In the **Cross-reference** pop-up box, set **Reference type:** to **Figure**, and set **Insert**
- 1324 **reference to:** to **Entire caption**.
- 1325 • Scroll through the **For which caption:** list and select the appropriate figure that is
- 1326 being cited.
- 1327 • Select the **Insert** button.
- 1328 • Change the font color of the inserted figure caption to **Blue**.

1329 **Cross-References—How to Update**

1330 When cited references are added or removed from the References section, all cross-references

1331 in the document need to be updated. Similarly, when sections are added or removed, section

1332 numbers will change and all cross-references to sections, tables, and figures will need to be

1333 updated. These can be updated individually or all at once.

1334 • To update all cross-references at once, select the entire document <Ctrl+A>, then press
1335 **F9**. It is important to confirm that all cross-references were updated and go to the
1336 correct cited reference, section, table, or figure by selecting each updated cross-
1337 reference <Ctrl+click>.

1338 To update an individual cross-reference, right-click on the specific cross-reference and select
1339 **Update Field**. Confirm the cross-reference was updated and goes to the correct cited
1340 reference, section, table, or figure by selecting on the updated cross-reference <Ctrl+click>.

1341 **Navigation Shortcuts**

1342 Use the **Navigation Pane** to quickly jump to a section or specific text in the document rather
1343 than scrolling through the pages.

1344 On the **View** tab, check the **Navigation Pane** box to display it.

- 1345 • **Search for a section:**
 - 1346 ○ **Headings:** Click on a section title to jump to that section.
 - 1347 ○ **Pages:** Scroll through page thumbnail images to quickly find a table, image, or
 - 1348 other content when a section title is unknown and a visual clue is needed. Click
 - 1349 on the thumbnail image of interest to jump to that page.
- 1350 • **Search for a word or phrase:** Type the text in the **Search document** at the top of
1351 the Navigation Pane.
 - 1352 ○ **Headings:** Section titles containing the typed text are highlighted in yellow.
 - 1353 ○ **Pages:** Thumbnail images are displayed for only those pages containing the
 - 1354 typed text.
 - 1355 ○ **Results:** Each individual search result is displayed with some of its surrounding
 - 1356 text.

1357 **Check Accessibility**

1358 Use Microsoft's **Accessibility Assistant** to identify and fix common accessibility issues that
1359 make it difficult for blind or low vision people. The **Status Bar** at the bottom of the window will
1360 display **Accessibility: Investigate** if there is an accessibility issue.

1361 On the **Review** tab, select **Check Accessibility** to open the Accessibility pane on the right.
1362 Issues and suggested resolutions are listed in the Accessibility pane.

- 1363 • **Hard-to-read text contrast:** If the text color is hard to see. Use the styles listed in the
1364 **Styles** gallery as they do not cause accessibility issues.
- 1365 • **Missing Alt Text:** A short 1-2 sentence description is needed to describe what the
1366 image is for someone who is blind or has low vision.
- 1367 • **Use of merged or split cells:** Cells should not be merged or split as it makes it difficult
1368 for screen readers to convey the information.

1369 **Inclusive Language**

- 1370 • Review the document to confirm that terms used in the document meet NENA's [Use of](#)
1371 [Inclusive Language Policy](#), found in the NENA Workspace Administrative Procedure,
1372 Templates & Training Material.

